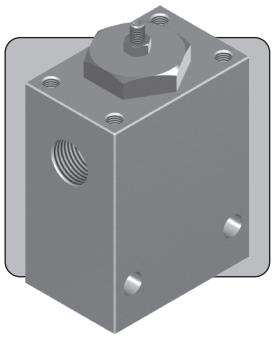
PCFR50

FLOW REGULATOR

1/2" 3/8"

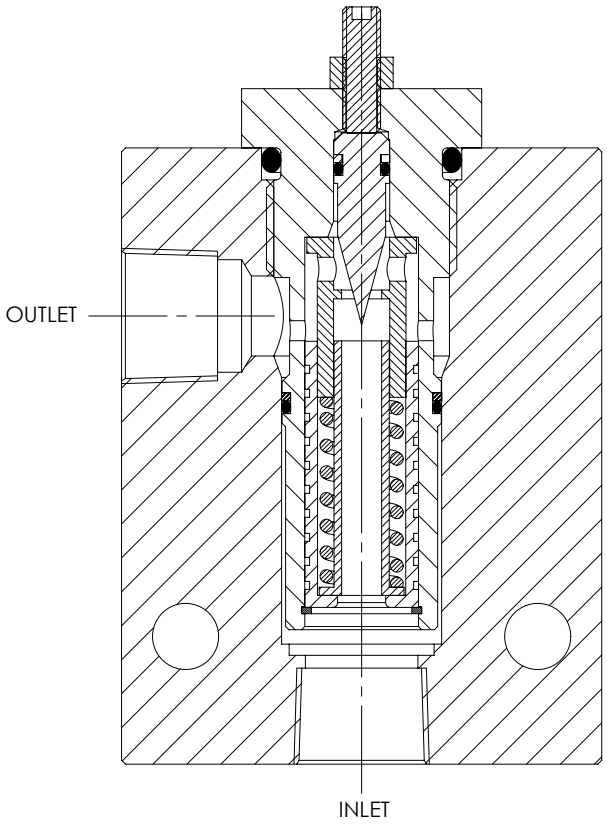
UP TO 690 BAR
10,000 PSI



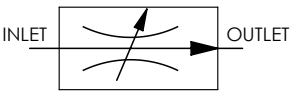
The PCFR50 is a pressure compensated flow control valve. Unlike a standard metering valve, the PCFR50 provides a relatively constant flow in one direction, regardless of the pressure differential across the valve, and is ideal for providing constant speed control of functions that have variable loads.

In reverse, the valve will open to allow non-metered reverse flow once a differential pressure of approximately 7 bar (102 psi) or 2 bar (29 psi) for LF versions is exceeded, and below this pressure will provide non-pressure compensated metered flow. This valve does not provide tight shut-off when closed.

- Pressure compensated flow control valve
- Adjustable flow setting with locking feature
- Ideal for uni-directional speed control with variable loads or supply pressures
- Stainless steel construction
- Suitable for use with mineral oils and water glycols and many other media, contact us for advice
- Subsea variant available, contact us for advice
- Various porting options available, including a cartridge version for use in manifolds



Specifications

BASIC MODEL NUMBER	PCFR50
SYMBOL	
MAX WORKING PRESSURE (LIQUID)	690 bar (10,000 psi)
MAXIMUM DIFFERENTIAL PRESSURE	414 bar (6,000 psi)
MINIMUM DIFFERENTIAL PRESSURE	15 bar (218 psi) When operated below this differential pressure, flow rate will not be pressure compensated
REGULATED FLOW SETTING RANGE	5 - 45 l/min (1.3 - 11.9 USgpm) For lower flows, see LF version in Product Selector opposite
FLUID	Liquids only See materials section
TEMPERATURE RANGE	See Product Selector opposite and Technical Data section
PORT SIZE	1/2" or 3/8" See Product Selector opposite
WEIGHT	Standard 2.8 kg (6.2 lb) PCFR50C 0.35 kg (0.8 lb)

Specifications may change without notice

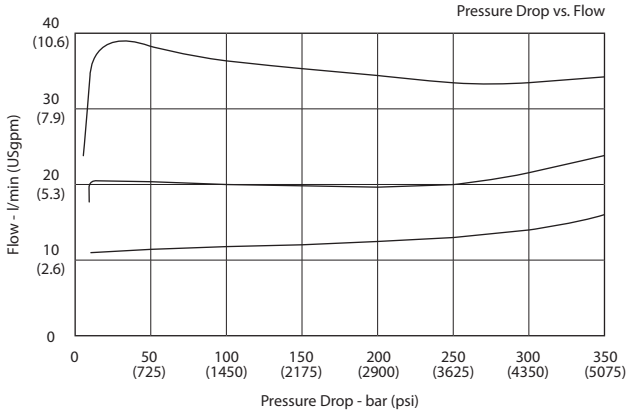
Materials

Externally Exposed Parts: 316 and 304 stainless steel.
Internally Wetted Parts: 316, 302, 431 and 440C stainless steel and acetal for the standard valve. 316, 302, 17-4 PH, acetal and aluminium bronze for the water duty version.

The standard valve is designed for use with non-corrosive liquids only, such as mineral oil and water glycol. A water duty version is available for use in plain water systems. Contact us for advice where use with more corrosive fluids is required.

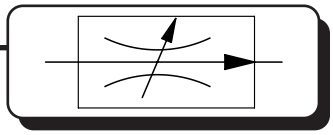
⚠ The standard valve has Viton® seals. Further seal options are available via the Product Selector. Compatibility with the working fluid at the operating temperature must be considered.

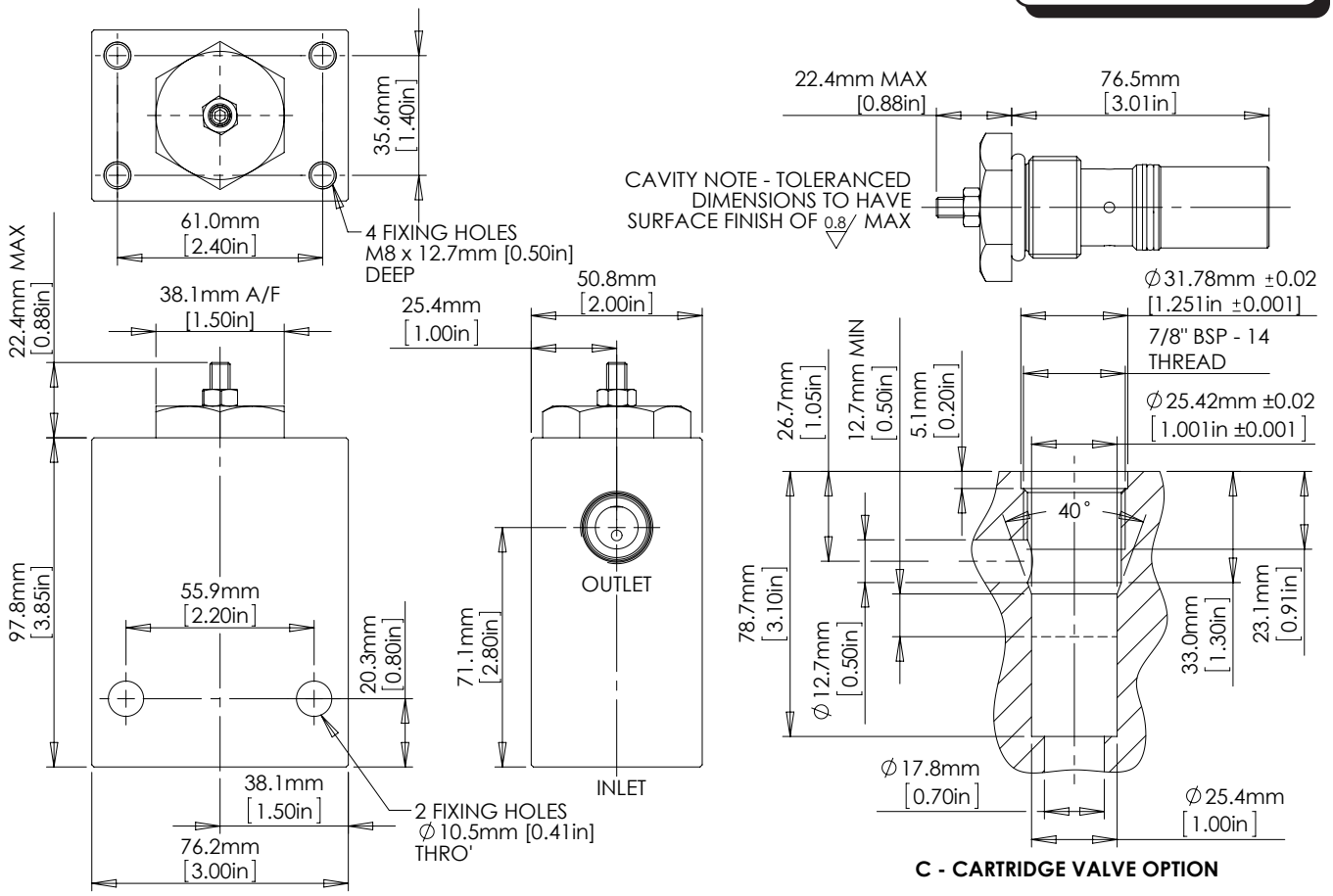
Typical Performance



Typical performance based on water

Installation Diagram





C - CARTRIDGE VALVE OPTION

Product Selector

PCFR50 P - LF - NBR - 10K

VALVE TYPE	PORTING OPTIONS	VALVE OPTIONS	SEAL OPTIONS	MAX WORKING PRESSURE
		Leave blank if not required	Leave blank for Viton® seals -10˚C to +120˚C	
PCFR50	P 1/2" BSPP female	LF Low flow 2 - 15 L/Min (0.5 - 4.0 USgpm)	NBR Nitrile seals -30˚C to +105˚C	10K 690 bar (10,000 psi) Liquid
	N 1/2" NPT female	WD Water Duty	EP EPDM seals -45˚C to +120˚C Only available with the WD option	
	/37P 3/8" BSPP female	Any combination of options is allowable	KLZ Kalrez® seals -25˚C to +120˚C	
	/37N 3/8" NPT female		HNBR Hydrogenated Nitrile seals -20˚C to +120˚C	
	C Cartridge valve		Further seal options are available on request	

Repair and seal kits are available on request

For a seal kit add SK at the end of the model code and for a repair kit add RK